Water Treatment and Purification Technology

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Introduction to Water Treatment

Water treatment is the process of removing harmful impurities such as dirt, germs, chemicals and excess minerals from raw water so that it becomes safe to drink or use. Raw water from rivers, lakes, reservoirs and wells may contain suspended particles, bacteria, viruses, parasites, dissolved salts and pollutants. Unsafe water causes diseases such as cholera, typhoid, dysentery and hepatitis A. The World Health Organization notes that safe drinking water is essential for health, and access to clean water is part of the United Nations Sustainable Development Goal 6. In India, the Bureau of Indian Standards sets the drinking water quality standard IS 10500.

What Is Reverse Osmosis?

Reverse osmosis (RO) is a water purification technology that removes dissolved salts, metals and other impurities by forcing water through a semi-permeable membrane under high pressure. To understand it, recall osmosis: in natural osmosis, water moves through a semi-permeable membrane from a dilute solution to a concentrated solution. In reverse osmosis, pressure higher than the natural osmotic pressure is applied to the salty side, which pushes water molecules in the opposite direction, from the concentrated side to the pure side. The membrane has extremely tiny pores of about 0.0001 micron that let water molecules pass but block most dissolved salts, bacteria, viruses and heavy metals. The water that passes is called permeate or purified water, and the concentrated salty water that is left behind is called reject or brine. A typical household RO system also includes a sediment filter, an activated carbon filter, a pump, a storage tank and often a UV lamp. RO is useful where water has high Total Dissolved Solids (TDS), but it wastes a significant amount of water, often two to three litres for each litre purified, and it can remove useful minerals, so many purifiers add a mineral cartridge. RO is also the key technology in desalination plants that convert seawater into drinking water.

The Main Stages of Municipal Water Treatment

Large water treatment plants follow a series of stages. The first stage is screening and intake: raw water passes through screens that remove leaves, fish and debris. In coagulation, chemicals such as alum (aluminium sulphate) or ferric chloride are added, which make tiny suspended particles stick together. In flocculation, gentle stirring helps these particles form larger clumps called flocs. In sedimentation, the water rests in large tanks and the heavy flocs settle to the bottom as sludge. In filtration, water passes through layers of sand, gravel and sometimes activated carbon, which remove remaining fine particles. In disinfection, chlorine or other disinfectants kill disease-causing microorganisms. Finally, the treated water is stored in tanks and distributed through pipes. Additional steps may include pH adjustment, fluoridation and aeration.

Why Is Chlorine Added to Water?

Chlorine is added to water as a disinfectant. It kills or inactivates bacteria, viruses and other microorganisms that cause waterborne diseases such as cholera and typhoid. Chlorination is cheap, effective and, importantly, leaves behind a small amount of residual chlorine that continues to protect the water as it travels through long pipelines to homes. The first continuous large-scale use of chlorine to disinfect public drinking water in the United States was in Jersey City in 1908, and chlorination has since saved millions of lives. The amount must be carefully controlled, because too much chlorine causes a bad taste and can form harmful by-products. Alternatives and supplements include ozone, chloramine and ultraviolet (UV) light.

Household Purification Methods

Boiling water for at least one minute kills germs. Candle filters and ceramic filters remove particles and some bacteria. Activated carbon filters remove chlorine, odour and some chemicals. UV purifiers use ultraviolet light to destroy microorganisms but do not remove dissolved salts. Ultrafiltration (UF) membranes remove bacteria and viruses without electricity. Choosing a purifier depends on the TDS level and the water source: RO suits high-TDS borewell water, while UV or UF suits low-TDS municipal water.

Desalination and Wastewater Treatment

Desalination converts seawater into fresh water using RO or distillation. Wastewater, or sewage, is also treated at sewage treatment plants through primary, secondary (biological) and tertiary stages, and treated water can be reused for industry, gardening and construction. Zero Liquid Discharge systems recycle all wastewater in industries.

Water Technology in India and Chennai

Chennai, which faces periodic water scarcity, has pioneered desalination in India. The Minjur seawater desalination plant, commissioned in 2010 with a capacity of 100 million litres per day, and the Nemmeli plant, which began operation in 2013 with the same capacity, supply drinking water to the city. A larger plant at Perur has also been planned. The Jal Jeevan Mission, launched in August 2019, aims to supply tap water to every rural household. Chennai Metropolitan Water Supply and Sewerage Board (CMWSSB) operates treatment plants at Kilpauk, Puzhal and Chembarambakkam. The Namami Gange programme treats sewage to clean the Ganga river.

Challenges and Future

Challenges include pollution of sources, groundwater contamination by fluoride, arsenic and nitrate, high energy use of desalination, disposal of brine and water wastage by RO. The future includes solar-powered desalination, graphene and nano membranes, smart sensors for water quality monitoring and wider reuse of treated wastewater.

Quick Facts

- Reverse osmosis forces water through a semi-permeable membrane under pressure to remove dissolved salts and impurities.
- Municipal treatment stages are screening, coagulation, flocculation, sedimentation, filtration and disinfection.
- Chlorine is added to kill disease-causing germs and to leave residual protection in pipes.
- Chennai's Minjur (2010) and Nemmeli (2013) plants each desalinate 100 million litres per day.
- The Jal Jeevan Mission was launched in August 2019 to provide tap water to rural homes.